

WASP-43b

R-0619

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_180222 · created 2026-07-25T11:34:09+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458171.8437 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.057
Transit depth (Rp/Rs)^2	3.89 +/- 2.24 [%]
Orbital Inclination (inc)	86.94 +/- 2.79
Airmass coefficient 1 (a1)	0.958 +/- 0.059
Airmass coefficient 2 (a2)	0.02 +/- 0.12
Scatter in the residuals of the lightcurve fit is	6.08 %
Best Comparison Star	#3 - [364, 90]
Optimal Aperture	3.02
Optimal Annulus	6.76
Transit Duration (day)	0.0613 +/- 0.0045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.197 $\pm$ 0.057 vs 0.1594 (deviation 0.66 σ)
Duration fit vs published	0.0613 d vs 0.0512 d (deviation 2.23 σ)
Mid-time O–C	-3.3 min
Depth SNR	3.5
Residual scatter	6.08 %

Light curve

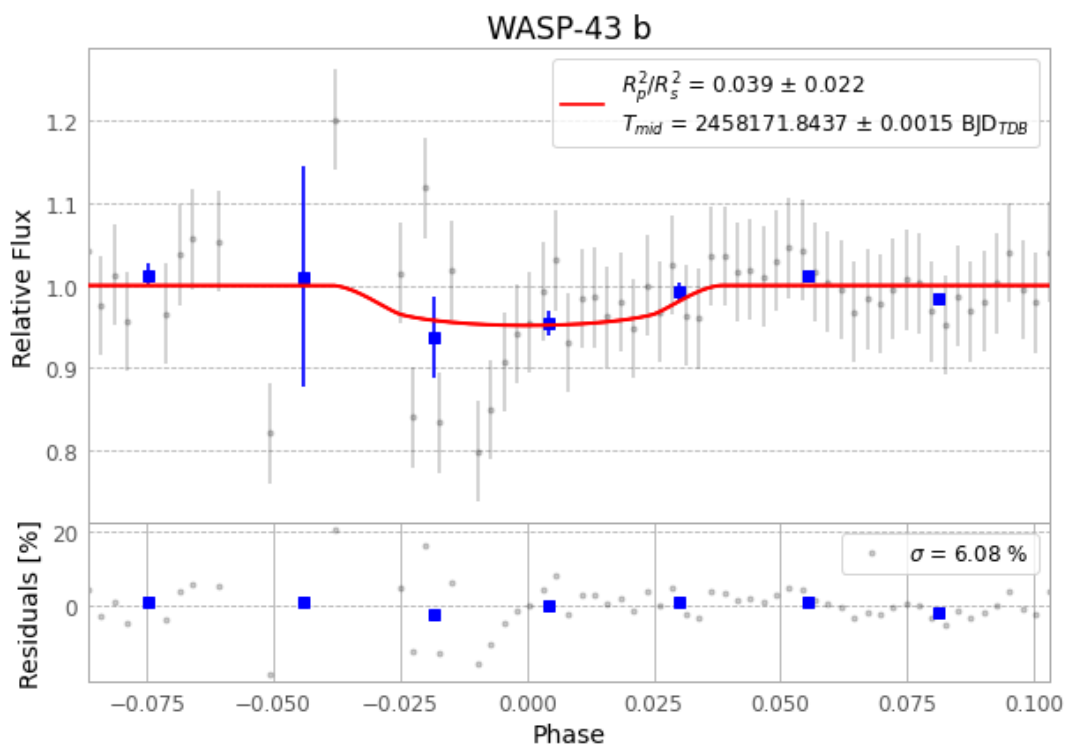


Figure 1 — FinalLightCurve_WASP-43 b_2018-02-21.png (SHA-256 278a884b16eb2b63...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [441, 284], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6eb10096725c6316...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

75 FITS frames (49 MB), WASP-43180222062414.FITS → WASP-43180222100612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-180222.log (78776ca2261a...), FinalLightCurve_WASP-43 b_2018-02-21.png (278a884b16eb...), FinalLightCurve_WASP-43 b_2018-02-21.csv (1e524bda1997...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 11:34Z.